SUPPLY CHAIN PERFORMANCE & OPERATIONS ANALYTICS

End-to-End Enterprise Intelligence: Procurement SLA, Multi-Echelon Inventory Health, Carrier Reliability & Commercial Profitability

Project Scope: Global Supply Chain & Commercial BI	**Tools:** Power BI, DAX, SQL, Python, Star Schema
Dataset Scale: 10 Tables, 24,000+ Transactions	**Positioning:** Executive Operations & Risk Analytics

1. EXECUTIVE SUMMARY

Modern global consumer electronics supply chains operate under intricate multi-echelon interdependencies, linking Tier 1 component vendors, advanced assembly plants, regional distribution hubs, multimodal freight carriers, and omnichannel retail networks. This analytics case study establishes a centralized **Supply Chain Operations & Financial Risk Control Tower** designed to diagnose bottlenecks, eliminate capital inefficiency, and maximize commercial margin capture across a global consumer hardware portfolio.

Key High-Level Metrics: Over the 24-month reporting period (2023–2024), the enterprise generated **$176.95M in Net Revenue** (Gross: $186.86M) and **$48.56M in Net Operating Profit** (27.44% net margin) across 187,130 units sold. Procurement operations committed **$78.13M across 2,200 POs** with an average lead time of **11.53 days** (exceeding the 10-day benchmark SLA). Logistics dispatches executed **7,500 shipments ($19.42M freight cost)** with a **7.64% delay exception rate (573 delayed trips)**. Inventory diagnostics revealed **$108.77M in buffer capital**, of which **$15.69M is locked in overstocked inventory positions** while 25 distribution center snapshots suffer from critical stockout vulnerabilities.

2. PROJECT POSITIONING & COMPLIANCE DISCLAIMER

Educational & Portfolio Project Notice: *This project utilizes a comprehensive synthetic consumer electronics dataset modeled after industry supply chain architectures. It is developed independently for advanced business intelligence, data modeling, and operations analytics portfolio demonstration. It does not represent actual operational, confidential, or financial data of Samsung Electronics.*

3. DATA AUDIT & DATA MODEL REFACTORING

A thorough audit of the source assets identified critical inconsistencies in legacy data modeling and business metric definitions:

- **Defective Inventory Value Metric:** The legacy DAX model calculated Inventory Value as SUM(fact_inventory[stock_level]), representing physical units rather than financial capitalization. Refactored to SUMX(fact_inventory, stock_level * RELATED(dim_product[unit_cost])), correcting the valuation from 160K units to $108.77M USD.
- **COGS vs Procurement Spend Confusion:** The legacy metric for Sales Cost incorrectly summed fact_procurement[total_cost]. Procurement spend and Sales Cost of Goods Sold (COGS) are distinct accounting concepts. Refactored to compute true COGS from fact_sales[total_cost] ($128.39M).
- **Temporal Disconnection:** Legacy fact tables (fact_procurement and fact_shipment) lacked active date relationships to dim_date, causing date slicers to malfunction. Established active star-schema foreign keys on order_date_key and ship_date_key.
- **Shipment Orphan Foreign Keys:** Identified 4,142 shipment records referencing unlisted product IDs and 5,295 records referencing unlisted customer IDs, which produced a 70.6% blank slice in legacy visuals. Documented ETL cleansing protocols to enforce referential integrity.

4. SUPPLY CHAIN CONTROL TOWER — CORE KPI SUMMARY

KPI Metric	Domain	Realized Value	Benchmark / SLA	Status / Assessment
Net Revenue	Commercial	$176.95M	$165.00M Target	Exceeding (+7.2%)
Operating Profit Margin %	Commercial	27.44%	25.00% Benchmark	Healthy (+244 bps)
Commercial Discount Rate	Commercial	5.31% ($9.92M)	< 5.00% Target	Elevated Margin Burn
Total Procurement Spend	Procurement	$78.13M	N/A (2,200 POs)	High Tier 1 Concentration
Average Vendor Lead Time	Procurement	11.53 Days	10.00 Days SLA	Lagging (+1.53 Days)
Supplier Quality Compliance	Procurement	96.63 / 100	95.00 Minimum	Compliant
Manufacturing Defect Rate	Production	0.629%	< 1.000% SLA	Optimal (24.1K units)
Total Inventory Valuation	Inventory	$108.77M	N/A (159.9K units)	Capital Imbalance
Overstocked Inventory Capital	Inventory	$15.69M	< $5.00M Target	Excess Holding Cost
On-Time Dispatch Rate (OTIF)	Logistics	92.36%	95.00% Target	Sub-optimal SLA
Logistics Delay Rate	Logistics	7.64% (573 trips)	< 5.00% Benchmark	Carrier Capacity Lag
Average Freight Trip Cost	Logistics	$2,589.02	$2,500.00 Target	Freight Inefficiency

5. MULTI-ECHELON OPERATIONAL DIAGNOSTICS & FINDINGS

A. Procurement & Vendor Sourcing Vulnerabilities

Analysis of 2,200 purchase orders across 7 major suppliers indicates severe spend concentration: the top 3 vendors (**BOE Technology at $11.49M, Samsung Vietnam at $11.40M, and TSMC at $11.33M**) absorb **43.8% of global spend**. Lead times show pronounced geographic variance: vendors located in China (BOE: 12.47 days), Japan (Sony: 12.42 days), and South Korea (SEC: 12.42 days) exhibit an 8-day delivery spread (P95 = 18 days) compared to Vietnam (9.39 days) and India (9.46 days). This delivery spread forces downstream assembly plants to maintain elevated safety stocks, increasing working capital requirements.

B. Inventory Buffer Health & Stockout Risk Analysis

A multi-echelon audit of 1,152 inventory snapshots across 6 production plants and distribution centers revealed a dual-sided structural imbalance:

- **Overstocked Positions ($15.69M):** 135 warehouse snapshots hold stock exceeding 2.5 times safety stock thresholds, primarily in mature tablet and accessory product lines, driving unnecessary storage and insurance carrying costs.
- **Critical Stockout Vulnerabilities ($1.68M at risk):** 25 warehouse snapshots operate below safety stock limits, specifically for high-velocity flagship smartphones (Galaxy S24 Ultra and Galaxy Z Fold5), risking unfulfilled customer demand during peak cycles.

C. Logistics SLA & Carrier Reliability Quadrant

Logistics analysis across 7,500 dispatches identified significant performance divergence among freight partners. **Maersk Line recorded the highest delay exception rate (10.39% / 87 delayed dispatches)**, followed by **DHL Express (8.01%)** and **DB Schenker (7.78%)**. Root-cause decomposition of 573 delayed shipments revealed that **Carrier Capacity shortages (15.71% / 90 incidents)**, **Documentation issues (13.61% / 78 incidents)**, and **Port Congestion (12.74% / 73 incidents)** account for over 42% of all delay exceptions.

D. Commercial Profitability & Discount Elasticity

Commercial evaluation demonstrated that premium smartphones represent **43.7% of total revenue** (Galaxy S24 Ultra: $34.49M, Galaxy Z Fold5: $27.68M). However, commercial discounting conceded **$9.92M in gross margin (5.31% discount rate)**. Channel analysis indicates that while Online marketplaces generate the largest volume ($73.24M net revenue / 27.51% margin), excessive promotional discounting on flagship products eroded margins without generating statistically significant volume acceleration.

6. STRATEGIC RECOMMENDATIONS & ACTION ROADMAP

Pillar	Observation & Root Cause	Financial & Operational Impact	Recommended Management Action
Procurement	Lead time volatility in East Asian suppliers (12.4 days vs 10 days SLA)	Forces excessive safety stock buffers, increasing working capital requirements	Renegotiate tier-1 vendor SLAs with dynamic lead time penalties
Inventory	$15.69M capital tied in excess stock while 25 DC positions risk stockout ($2.3M/yr) and delayed reorder point triggers	Overstocked mature SKUs increase storage costs; stockouts risk customer churn	Deploy automated multi-echelon inventory optimization algorithms and execute
Logistics	Maersk (10.39%) and DHL (8.01%) drive high delay exception rates; carrier capacity shortages (15.71%) and documentation errors (13.61%)	Customer dissatisfaction, SLA breaches and detention/demurrage expenses	Realign freight volumes away from underperforming carriers (e.g., C.H. Robinson (6.92% delay); implem
Commercial	Uncontrolled discounting on flagship SKUs (Galaxy S24 Ultra, Galaxy Z Fold5)	Conceded $9.92M in margin; price-inelastic premium products	Implement strict discount governance floors; replace cash disco

7. CONCLUSION & PORTFOLIO IMPACT

By refactoring the legacy project into a rigorous, multi-echelon business intelligence solution, this project demonstrates advanced proficiency across the complete data analytics lifecycle: from low-level data extraction, SQL schema querying, and Python statistical diagnostics to star-schema data modeling, production DAX development, and executive-level narrative storytelling.